

ECE 260 Continuous-Time Signals and Systems

- **Tutorial 1**

Complex Analysis & AI for MATLAB

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Catalog

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Signal & System in my research

A short bridge from course tools to research practice.

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Complex analysis assignment hints

Method, setup, traps, and checks without final answers.

4

AI for MATLAB

Use AI as a tutor and debugger, not as an answer machine.

Self Introduction & Tutorial Organization

Name: Ruilin Wang

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Bio:

Currently, I'm a PhD student supervised by Prof. Xiaodai Dong. I'm working on applying neural networks into wireless communication. In tutorials, my goal is to help you understand the problem setup, not just the final answer.

Tutorial schedule:

Location: **ELW 215**

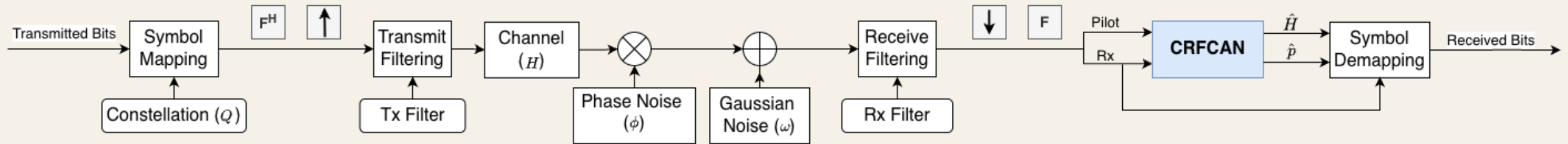
Time: Monday **1:30pm-2:20pm**

Optional Attendance & No Recordings Attendance for these tutorials is completely optional. If you feel comfortable with the lectures and assignments, you are not required to attend. However, please keep in mind that **these sessions will not be recorded**. If you are stuck on a concept, joining the session in person is the best way to get help.

How to Get the Most Out of These Sessions To make our time as productive as possible, I highly recommend emailing me at least 24 hours before our tutorial with any specific concepts, slides, or homework problems you find confusing. This allows me to prepare detailed explanations and tailor the session exactly to what you need.

Signal and System in My Research

Joint Estimation of Channel and Phase Noise in OFDM System



Where Signals & Systems appear in the OFDM chain

Signal view

Bits $\rightarrow$ QAM symbols $\rightarrow$ OFDM waveform.
Pilots/Rx samples carry H and phase-noise information.
We transform signals to reveal structure.

System view

Tx/Rx filters and channel H are system blocks.
Time domain: filtering/convolution.
Frequency domain: subcarrier gains.

Impairments + estimation

Phase noise ϕ rotates complex symbols.
Gaussian noise ω adds uncertainty.
CRFCAN estimates H and ϕ before demapping.

Simplified received-signal model: $y[n] = e^{j\phi[n]} (h * x)[n] + w[n]$

ECE 260 lens: complex exponentials · convolution · frequency response H · noise · estimation

A.1(C) — POLAR TO CARTESIAN

Task

Express $2 e^{j 7\pi/6}$ in the form $x + jy$.

Concept

Euler's relation:

$$e^{j\theta} = \cos \theta + j \sin \theta$$

Setup steps

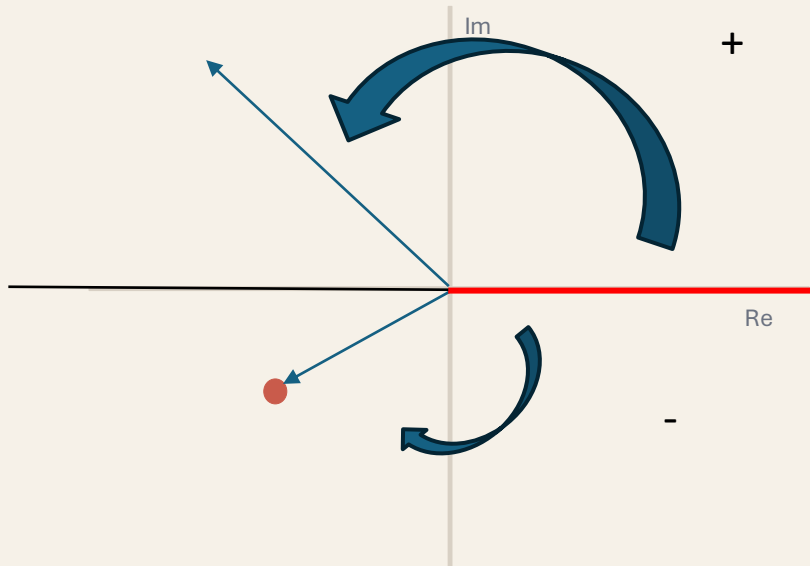
1. Identify r and θ .
2. Locate $\theta = 7\pi/6$ on the complex plane.
3. Decide the signs of $\cos \theta$ and $\sin \theta$ before calculating.

$$\begin{aligned} & 2 e^{j 7\pi/6} \\ &= 2 [\cos(7\pi/6) + j \sin(7\pi/6)] \end{aligned}$$

Stop here in tutorial.

Hint: $7\pi/6$ is in Quadrant III, so both components should be negative.

A.2(B,D) — CARTESIAN TO POLAR



Quadrant check first

Trap to name out loud

- Same tangent ratio can appear in different quadrants
- A negative real and negative imaginary part is not Quadrant I
- MATLAB angle(z) returns the principal value

$$r = \sqrt{x^2 + y^2}$$
$$\theta = \text{atan2}(y, x) \in (-\pi, \pi]$$

For each point

1. Plot the sign of x and y.
2. Compute magnitude.
3. Use atan2(y,x), not plain atan(y/x).
4. Convert θ to the requested angle convention.

A.3 — POWERS AND QUOTIENTS

$$(r e^{j\theta})^n = r^n e^{j n \theta}$$

$$\begin{aligned} & (r_1 e^{j\theta_1}) / (r_2 e^{j\theta_2}) \\ &= (r_1/r_2) e^{j(\theta_1 - \theta_2)} \end{aligned}$$

(f) $(1 + j)^{10}$

Hint sequence:

1. Convert $1+j$ to polar form.
2. Raise the magnitude to the 10th power.
3. Multiply the angle by 10.
4. Reduce the angle before converting back.

(g) $(1 + j)/(1 - j)$

Hint sequence:

1. Put numerator and denominator in polar form.
2. Divide magnitudes.
3. Subtract angles.
4. Convert only after simplifying.

Rule of thumb: Cartesian for addition/subtraction; polar for multiplication/division/powers.

A.4(B) AND A.6(B)

Arg quotient rule

Start with

$$z_1 = r_1 e^{j\theta_1}, \quad z_2 = r_2 e^{j\theta_2}$$

Then form z_1/z_2 and read the exponent. Be careful about principal-argument conventions.

Euler sine identity

Write both $e^{j\theta}$ and $e^{-j\theta}$.

Subtract the two expressions. The cosine terms cancel; the remaining term isolates $\sin \theta$.

Proof template:

rewrite $\rightarrow$ apply exponent laws $\rightarrow$ match the target identity

A.11(C,D) — ANALYTICITY OF A RATIONAL FUNCTION

$$f(z) = z / (z^4 - 16)$$

What not to forget

The equation $z^4 = 16$ has complex solutions. Do not stop after checking the real axis.

Decision path

1. Identify numerator $P(z)$ and denominator $Q(z)$.
2. Solve $Q(z)=0$ over the complex plane.
3. Those roots are candidate singularities.
4. State the domain by excluding them.

A rational function is analytic everywhere except at its poles

A.13(B) — CLEAR NEGATIVE POWERS FIRST

$$f(z) = z + 3 + 2z^{-1}$$

1

Single fraction

Multiply top and bottom by the highest power needed to remove z^{-1} .

2

Factor

Factor the numerator after standardization. Keep the denominator visible.

3

Read off

Only after simplification: zeros from $P(z)$, poles from $Q(z)$.

```
% After you derive coefficients by hand:  
num = [ ... ];  
den = [ ... ];  
roots(num);    % candidate zeros  
roots(den);    % candidate poles
```

Use the same workflow for every pole-zero problem.

1

Standardize

Write f as one fraction $P(z)/Q(z)$. Clear negative powers.

2

Factor

Factor P and Q into linear factors; use the quadratic formula for complex roots.

3

Cancel

Cancel common factors; these are removable singularities, not true poles/zeros.

4

Identify

Zeros come from simplified P ; poles come from simplified Q .

5

Order

Read the exponent on each factor $(z-a)^k$.

Use the exponent method in tutorials; know the definition for proofs.

Method A — exponent

From the simplified, factored form:

$$(z - a)^k$$

means $z = a$ has order k .

Use this most of the time.

Method B — definition

z_0 is a zero of order n if

$$f(z_0) = f'(z_0) = \dots = f^{(n-1)}(z_0) = 0, \\ \text{while } f^{(n)}(z_0) \neq 0.$$

For poles, apply this to $g(z) = 1/f(z)$.

Order 1 = simple. Order > 1 = repeated.

A.13(F) — SETUP WITHOUT THE FINAL TABLE

$$f(z) = z^2(z^2 - 1) / [(z^2 + 4z + 17/4)^2 (z^2 + 2z + 2)]$$

Numerator P(z)

$$z^2$$

→ root at $z = 0$ (order 2)

$$z^2 - 1 = (z - 1)(z + 1)$$

→ simple roots at $z = 1, -1$

Denominator Q(z)

$$(z^2 + 4z + 17/4)^2$$

roots $-2 \pm 0.5j$ → each order 2 (squared!)

$$z^2 + 2z + 2$$

roots $-1 \pm j$ → simple

TA tip: students often find roots correctly but miss the outer exponent on a quadratic factor.

How to write a prompt?

The way to use AI effectively and efficiently



1 Start simple, then refine

Ask once, inspect the failure, then add constraints.

2 Give stable context

Course, topic, notation, skill level, and allowed tools.

3 Use structure

Separate equation, background, task, and desired output with tags.

4 Force a sequence

Math idea → MATLAB functions
→ code → verification.

5 Add guardrails

Tell it not to guess; ask it to flag uncertain assumptions.

6 Standardize output

Request code blocks, comments, and a visualization section.

From **Prompting 101 | Code w/ Claude by Anthropic**

<https://www.youtube.com/watch?v=ysPbXH0LpIE&t=237s>

Weak prompt: "write MATLAB code". Strong prompt: "teach me the math, explain each function, then write verifiable code."

Ask in stages so you can catch mistakes before code becomes trusted.

- <stage 1>** Ask for the mathematical plan only. No MATLAB yet.
- <stage 2>** Ask which MATLAB functions are needed and what each input means.
- <stage 3>** Ask for code using your own coefficient vectors, with comments on every line.
- <stage 4>** Ask it to debug your error message or compare MATLAB roots with your hand factorization.

Good verification prompts

- "Check whether my coefficient vector matches this polynomial."
- "Explain why roots returns complex conjugate pairs here."
- "Help me plot zeros as o and poles as x without changing my math."

Never paste an unverified AI answer directly into an assignment.

The useful details are role, context, sequence, and output contract.

Prompt anatomy

Role: Signals and Systems professor + MATLAB expert.

Context: equation, background, notation.

Sequence: paper method before code.

Output: comments, code block, visualization.

Act as a university professor teaching Signals and Systems and as a MATLAB expert. Teach me; do not just give code.

```
<equation>  
[paste my expression here]  
</equation>
```

```
<background>  
I know basic complex arithmetic, but I need help with higher-order  
polynomials and pole-zero plots.  
</background>
```

Follow this order:

1. Explain the mathematical setup.
2. Explain MATLAB functions: roots, conv, zplane.
3. Write runnable code with detailed comments.
4. Add a <visualization> section explaining the plot.

Do not guess. If information is missing, say what is missing.

Questions?